Tracking the evolution of dynamic networks with indirect sampling

Antonio Fernández Anta
with Sergio Díaz-Aranda, Juan Marcos Ramírez, Leonhard Balduf, and many more…

El País, March 13th, 2020

España ya es el segundo país de la UE con más contagiados: 3.142 casos

How close is this to the real number?

EL PAÍS

www.elpais.com

EL PERIÓDICO GLOBAL

VIERNES 13 DE MARZO DE 2020 | Año XLV | Número 15.581 | EDICIÓN MADRID | Precio: 1,70 euros

ALEMANIA El servicio secreto vigila al ala más radical de la ultraderechista AfD P2

EE UU Detenidos 200 nar cartel heredero de El C

EL AVANCE DEL CORONAVIRUS DESAFÍA AL ESTADO

España, en emergencia

El Ejecutivo inyecta 14.000 millones en la economía y 3.800 millones en sanidad

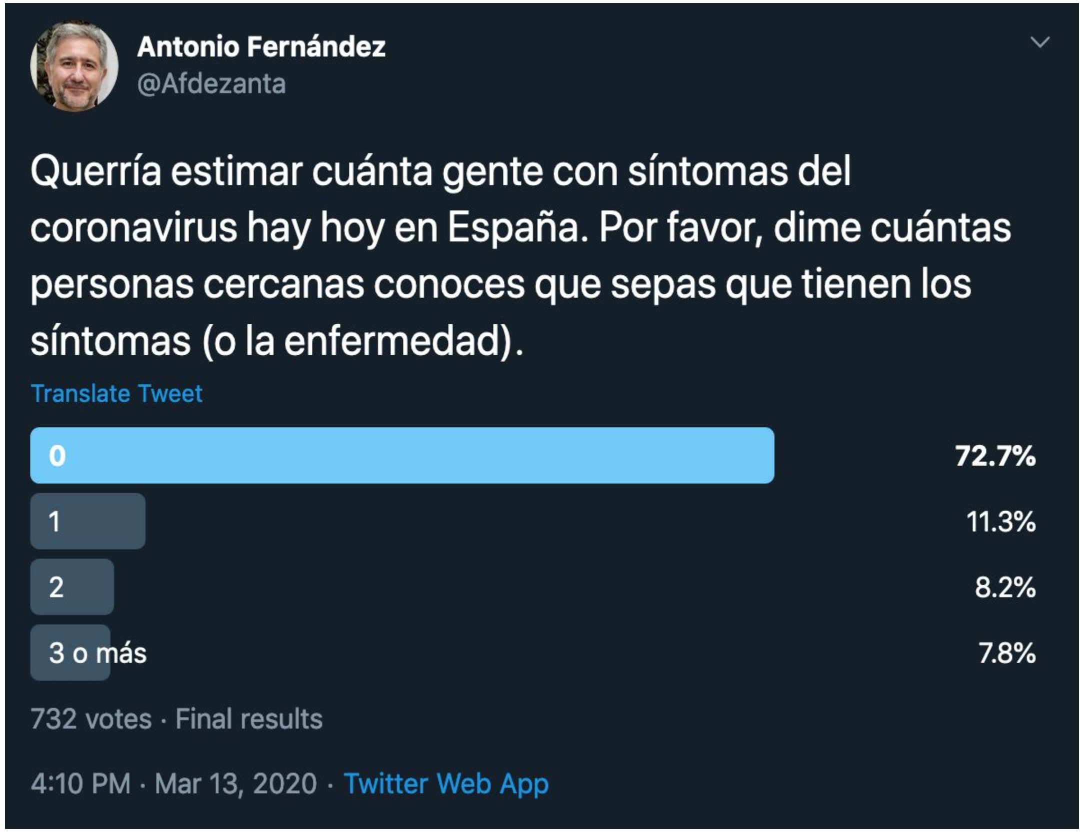
España ya es el segundo país de la UE con más contagiados: 3.142 casos

Sánchez propone un Presupuesto "extrasocial" y Arrimadas ofrece su voto

Casado critica p insuficiente el p apoyará en el C

COVID-19, March 14th, 2020

Official number of COVID-19 cases in Spain by March 14th, 2020:
6,391 ~ 0.0136% [OWID]



Antonio Fernández
@Afdezanta

Querría estimar cuánta gente con síntomas del coronavirus hay hoy en España. Por favor, dime cuántas personas cercanas conoces que sepas que tienen los síntomas (o la enfermedad).

Translate Tweet

0	72.7%
1	11.3%
2	8.2%
3 o más	7.8%

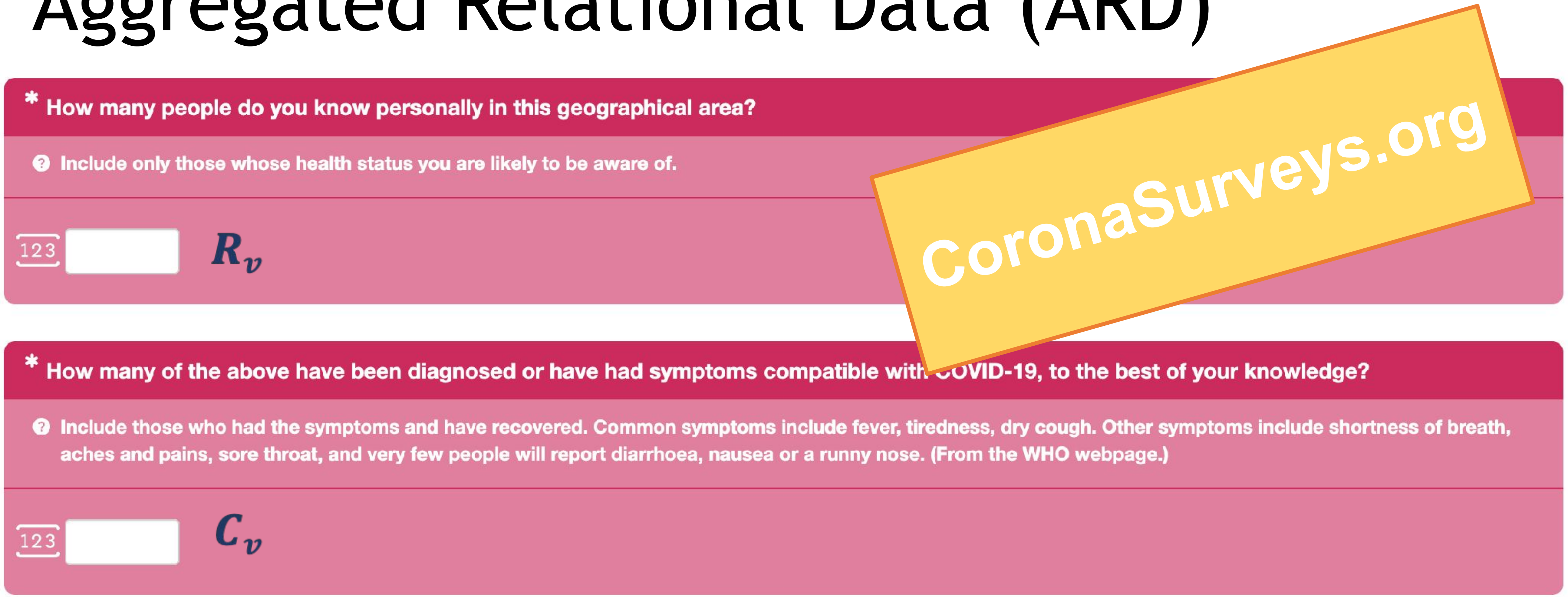
732 votes · Final results

4:10 PM · Mar 13, 2020 · Twitter Web App

732 responses
report 374 cases
& know
732 x 50 = 36,600 persons
(Dunbar # of ~50 friends)
▼
480,000 cases ~ 1%

14 days onset to death
1.38% Case Fatality Ratio
5,982 deaths (March 28th)
▼
433,478 cases ~ 0.92%

Aggregated Relational Data (ARD)



Aggregated Relational Data (ARD): Data collected from a survey of indirect questions:

- Privacy is preserved
- Each response reports the status of many individuals
- Responses can be biased

Network Scale-up Metod (NSUM)

* How many people do you know personally in this geographical area?

Include only those whose health status you are likely to be aware of.

123 R_v

* How many of the above have been diagnosed or have had symptoms compatible with COVID-19, to the best of your knowledge?

Include those who had the symptoms and have recovered. Common symptoms include fever, tiredness, dry cough. Other symptoms include shortness of breath, aches and pains, sore throat, and very few people will report diarrhoea, nausea or a runny nose. (From the WHO webpage.)

123 C_v

The prevalence ρ is estimated with the Network Scale-up Method (NSUM) [Bernard et al, 1991; Killworth et al, 1998]:

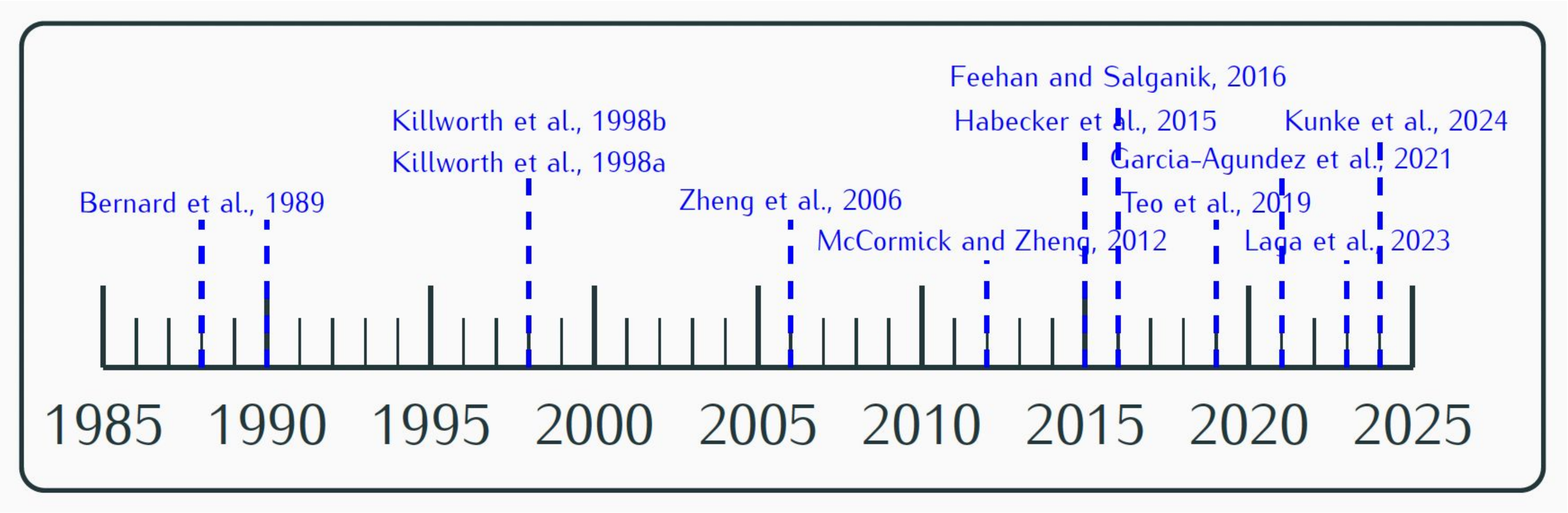
Ratio of Sums (RoS): $$\hat{\rho}_{\mathrm{RoS}}(I) \triangleq \frac{\sum_{v \in S} C_v}{\sum_{v \in S} R_v}$$

Mean of Ratios (MoR): $$\hat{\rho}_{\mathrm{MoR}}(I) \triangleq \frac{1}{|S|} \sum_{v \in S} \frac{C_v}{R_v}$$

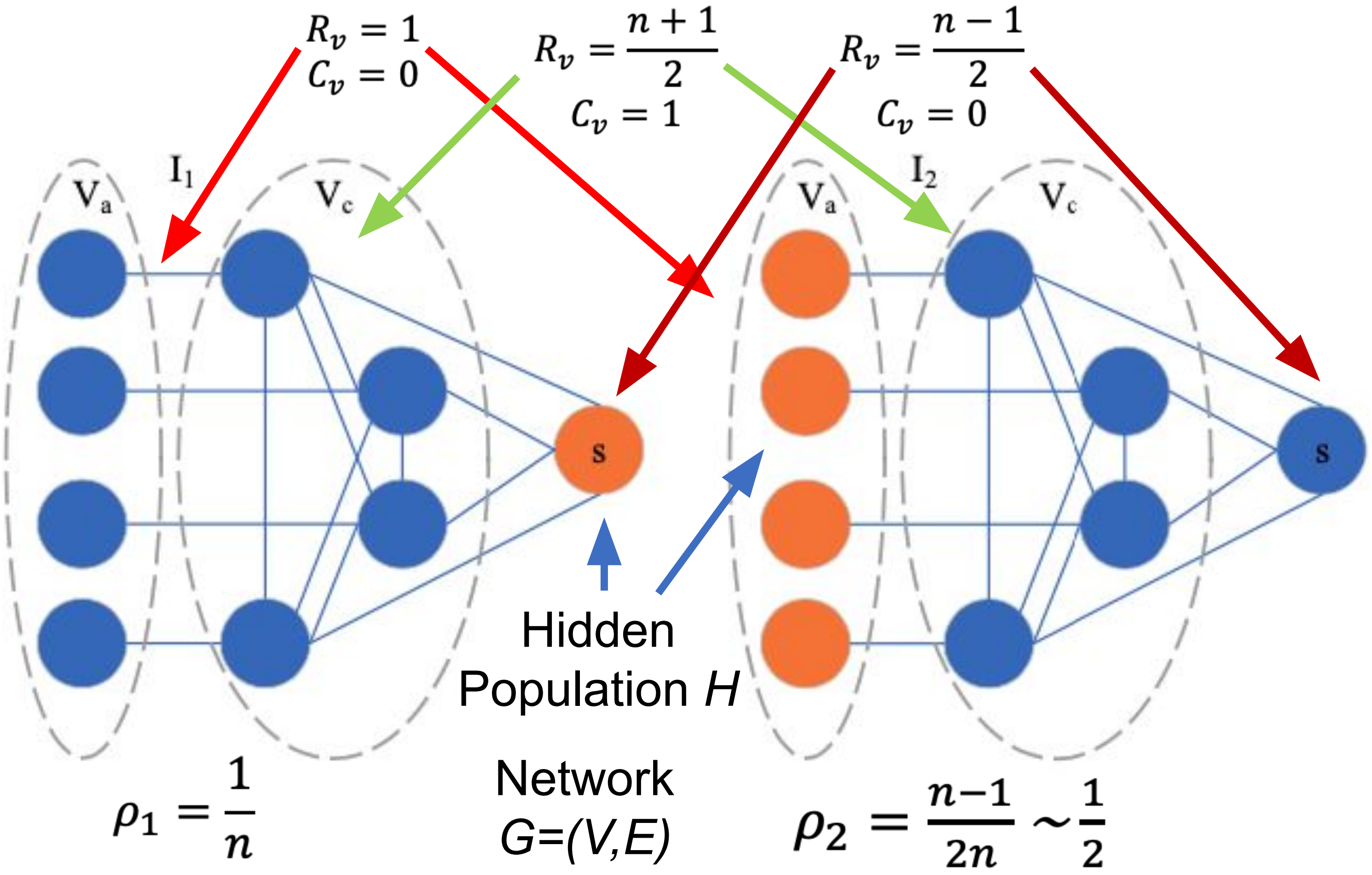
From ARD to Estimates: NSUM

ARD has been collected via indirect surveys to estimate:

- Casualties in an earthquake [Bernard et al. 1989]
- Prevalence of drug use [Salganik et al. 2010]
- Female sex workers conditions [Jing et al. 2018]
- Prevalence of HIV [Teo et al. 2019]
- Prevalence of COVID-19 [Garcia-Agundez et al. 2021]



How Bad Can NSUM Estimates Be?



Thm: Any deterministic NSUM estimation of the prevalence ρ that uses only the set

$$\{(R_v, C_v), \forall v \in V\}$$

has error

$$\mathcal{E} = \max\left(\frac{\rho}{\hat{\rho}}, \frac{\hat{\rho}}{\rho}\right) \geq \sqrt{\frac{n-1}{2}}$$

Analytical Study of NSUM

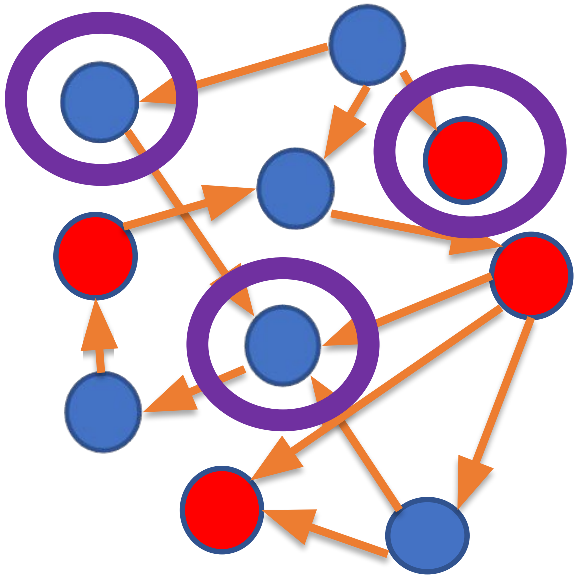
Analytical bounds:

- NSUM estimations can be very bad

Random Networks

Network construction:

1. Hidden population H is fixed
2. Vertex v selects an in-degree $R_v \geq 1$ from a degree distribution P_{deg}
3. Vertex v selects R_v vertices uniformly at random from $V - \{v\}$ to be its set of in-neighbors



Sampling:

1. A set S of m vertices from V is selected uniformly at random
2. Each vertex $v \in S$ reports

R_v (in-degree)

C_v (# of in-neighbors in H)

Estimators [Bernard et al, 1991; Killworth et al, 1998]:

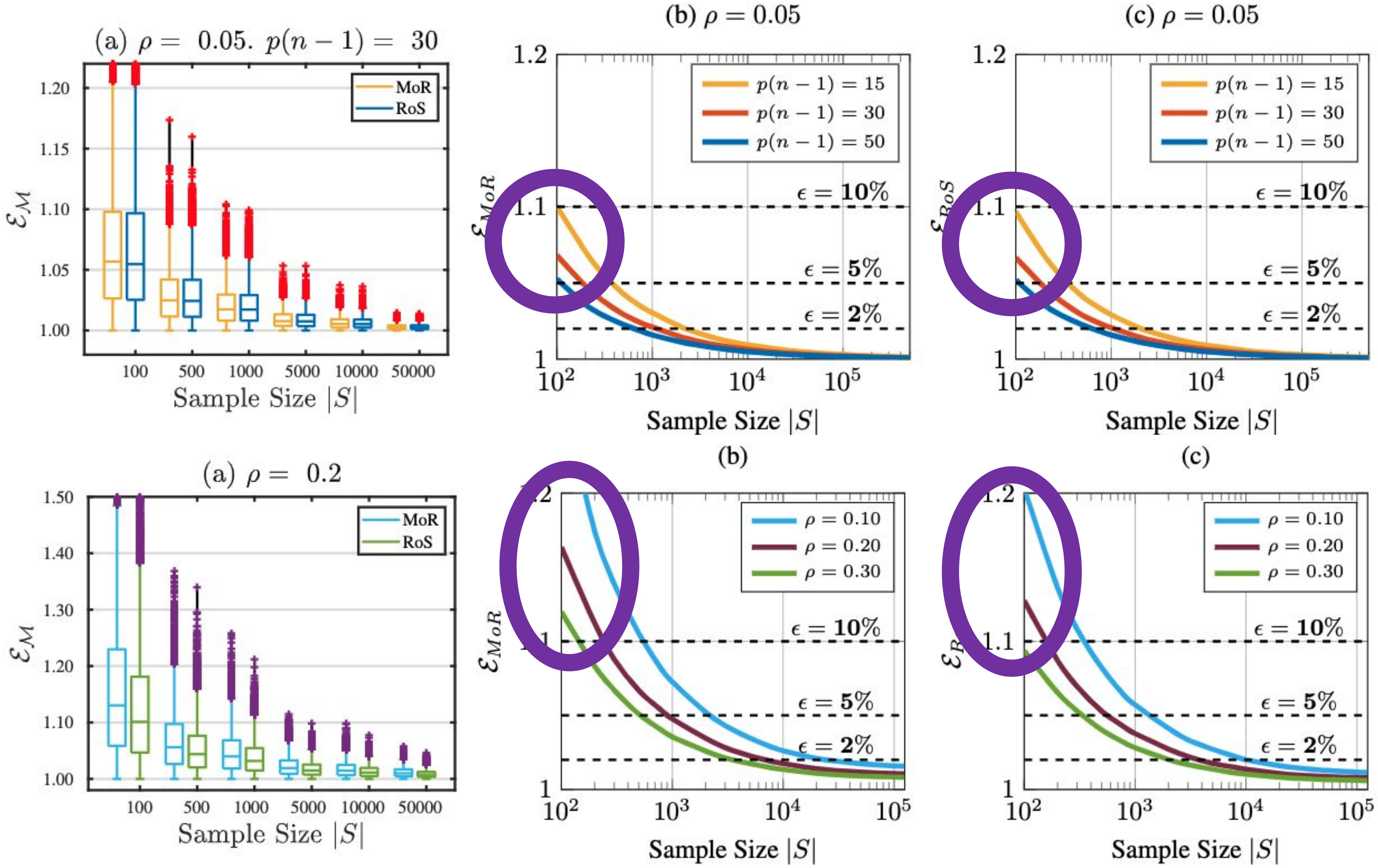
Ratio of Sums (RoS):

$$\hat{\rho}_{\mathrm{RoS}}(I) \triangleq \frac{\sum_{v\in S} C_v}{\sum_{v\in S} R_v}$$

Mean of Ratios (MoR):

$$\hat{\rho}_{\mathrm{MoR}}(I) \triangleq \frac{1}{|S|}\sum_{v\in S}\frac{C_v}{R_v}$$

Performance of MoR and RoS



Erdős–Rényi
$n = 10^6$

Scale-free
$n = 250{,}000$

Analytical Study of NSUM

Analytical bounds:

- NSUM estimates can be very bad
- If the network is "random" the NSUM estimates are good

Error Bounds for MoR and RoS

Theorem: *Consider an instance $I = (G, H)$ where G is a random network. If $S \subseteq V$ is sampled uniformly at random, with $m = |S|$, then for any $\beta = 1 + \varepsilon > 1$, for $R_S = \sum_{v \in S} R_v$,*

1. *The* RoS *estimator has the following error bound*

$$\Pr\left[\mathcal{E}_{\mathrm{RoS}} > \beta\right] \leq \sum_{R} \left(\left(\frac{e^{\beta - 1}}{\beta^{\beta}} \right)^{R\rho} + \left(\frac{e^{\frac{1}{\beta} - 1}}{\beta^{-1/\beta}} \right)^{R\rho} \right) \Pr\left[R_S = R\right]$$

2. *Both* MoR and RoS *estimators have the following error bound*

$$\Pr\left[\mathcal{E} > \beta\right] \leq \left(\frac{e^{\beta - 1}}{\beta^{\beta}} \right)^{m\rho} + \left(\frac{e^{\frac{1}{\beta} - 1}}{\beta^{-1/\beta}} \right)^{m\rho}$$

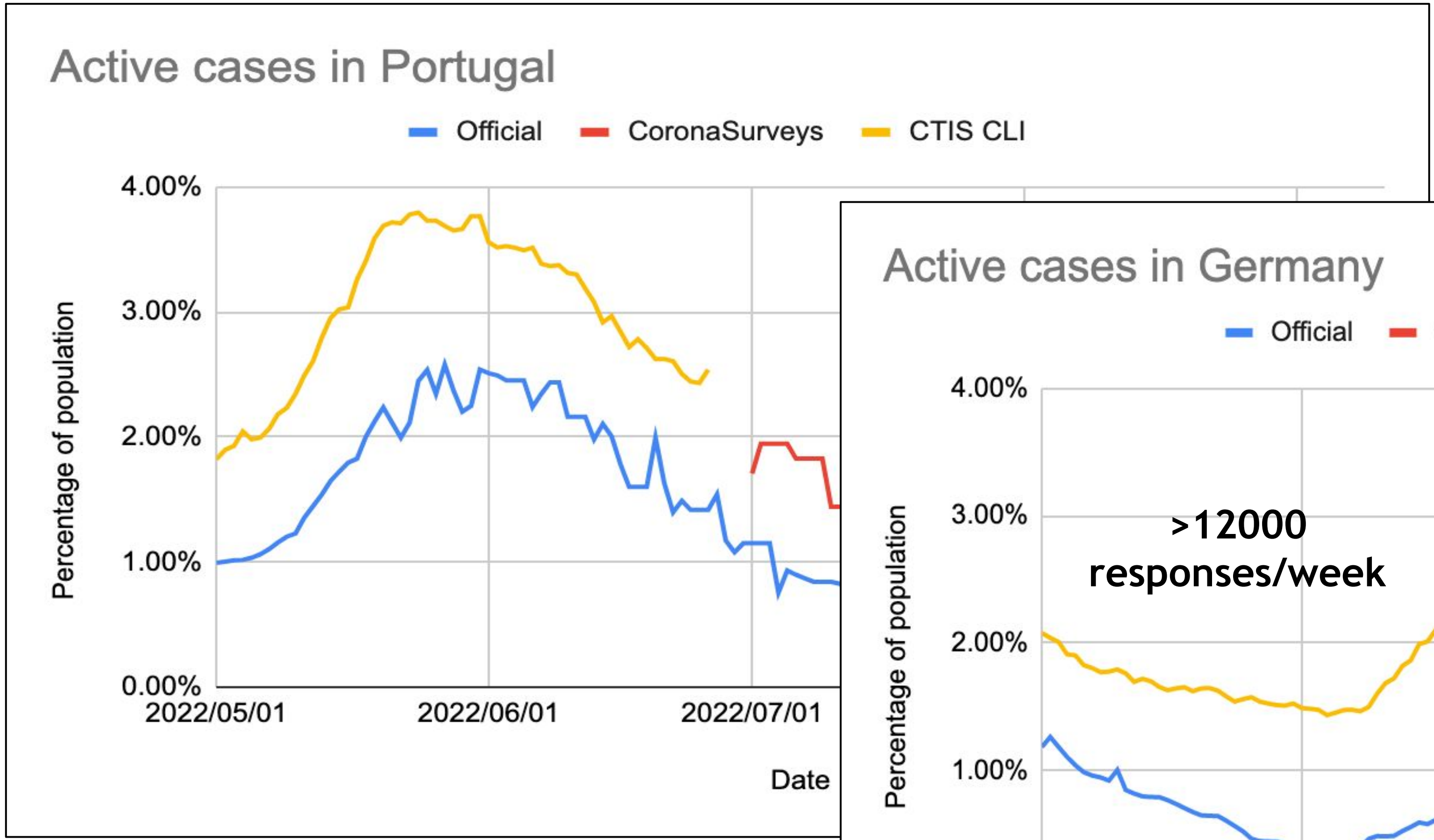
3. *Error bound β can be achieved with MoR and RoS with high probability $(1 - 1/n)$ using logarithmic sample size $m = |S| = O(\log n)$*

Analytical Study of NSUM

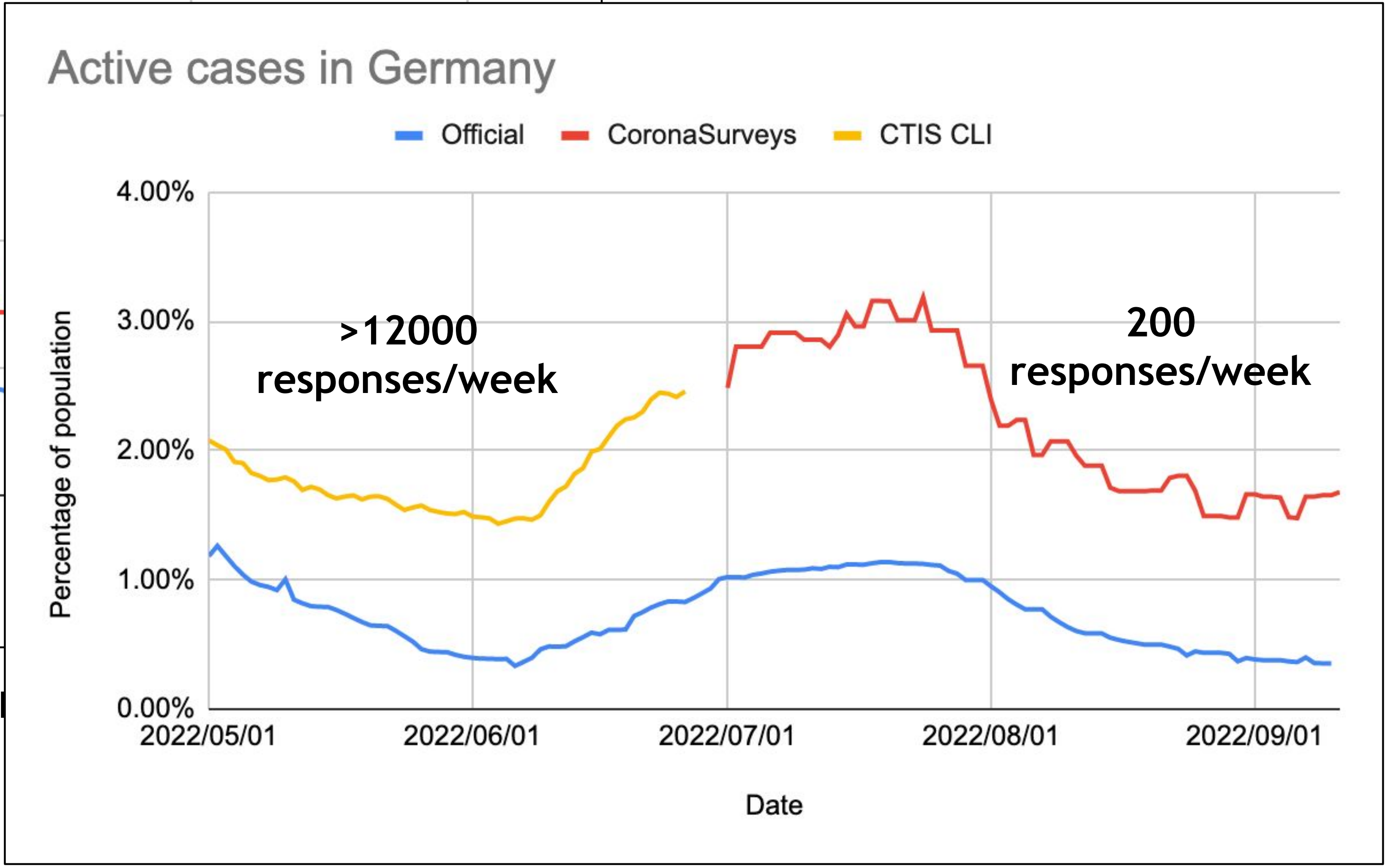
Analytical bounds:

- NSUM estimates can be very bad
- If the network is “random” the NSUM estimates are good
- The error can be bounded analytically

Monitoring COVID-19 Active Cases



Official: 10 x new cases, from official data [OWID]



CTIS CLI: fever & cough or difficulty br
COVID-19 Trends and Impact Survey

Temporal Social Network

Population V

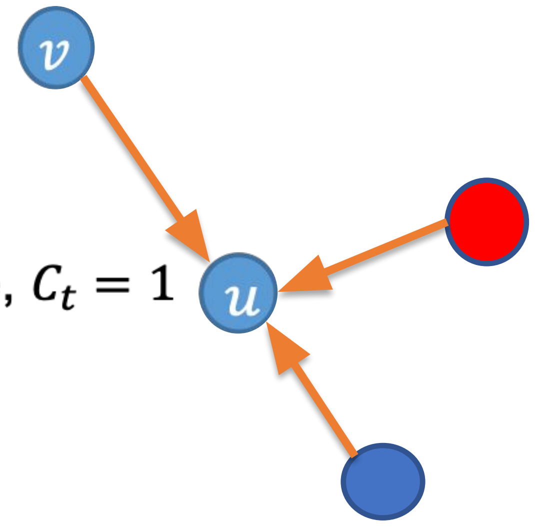
At time t:

- Hidden population H_t, and prevalence $\rho_t = \frac{|H_t|}{|N|}$
- Networks $G_t = (V, E_t)$, where $(v, u) \in E_t$ if u knows whether v is infected

Assumptions:

- In-degree distribution of all G_t have the same mean μ and variance σ^2 (supported empirically [Dunbar 2010])
- If $(v, u) \in E_t$, $\Pr[v \in H_t]$ does not depend on the in-degree of u in G_t

In this case, $C_t = 1$



Sampling:

- Select a node u at time t from G_t uniformly at random
- C_t is the number of hidden in-neighbors of u (random variable)

Prevalence Trend Estimation

Theorem: $E(C_t) = \mu \cdot \rho_t$

I.e.,

- The expectation of the indirect response C_t is proportional to the prevalence ρ_t we wish to estimate
- The time series $E(C_t)$ is proportional to the time series of ρ_t, with μ as the constant of proportionality
- The trend of ρ_t can be estimated without knowing μ
- If precise ρ_t values are needed, μ can be estimated (once) from reliable data

Analytical Study of NSUM

Analytical bounds:

- NSUM estimates can be very bad
- If the network is “random” the NSUM estimates are good
- The error can be bounded analytically

Díaz-Aranda, Sergio, et al. "Error bounds for the network scale-up method." *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 2*. 2025.

ARD to monitor social evolution:

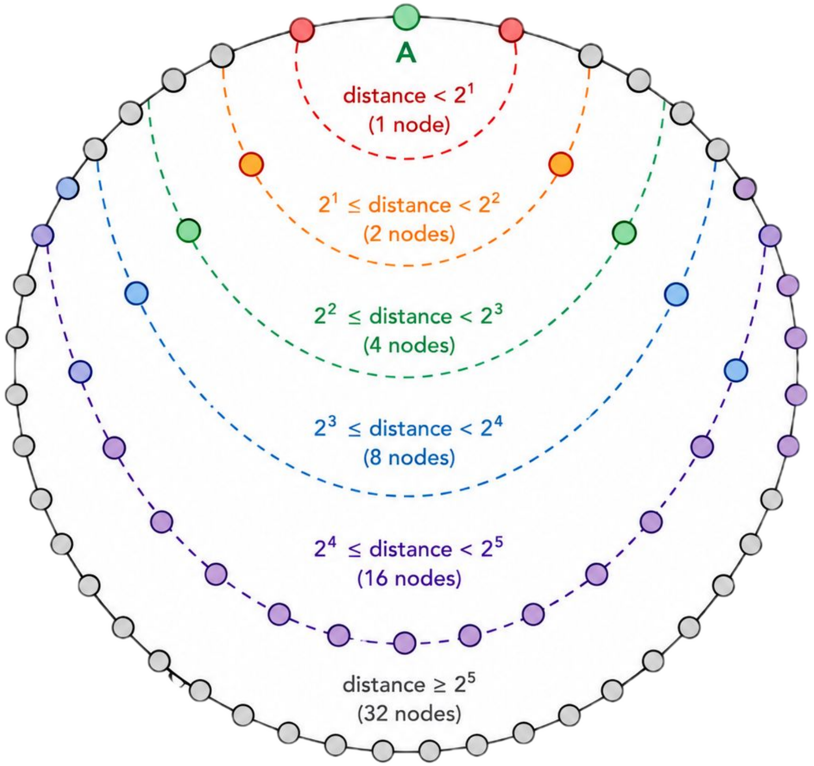
- Social trends can be followed with ARD
- Indirect reporting has less error than direct
- Smoothing improves the estimates

Srivastava, Ajitesh, et al. "Nowcasting temporal trends using indirect surveys." *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 38. No. 20. 2024.

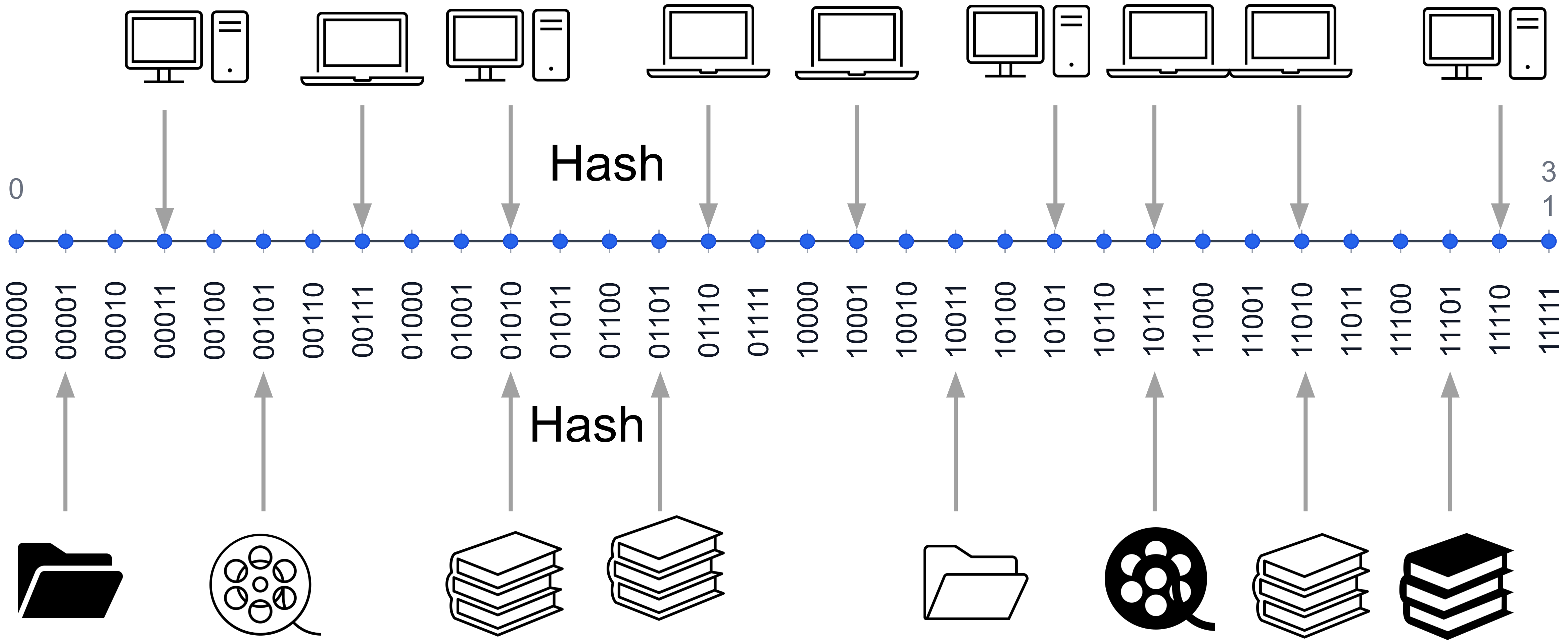
Application to Structured P2P Networks

Structured P2P networks:

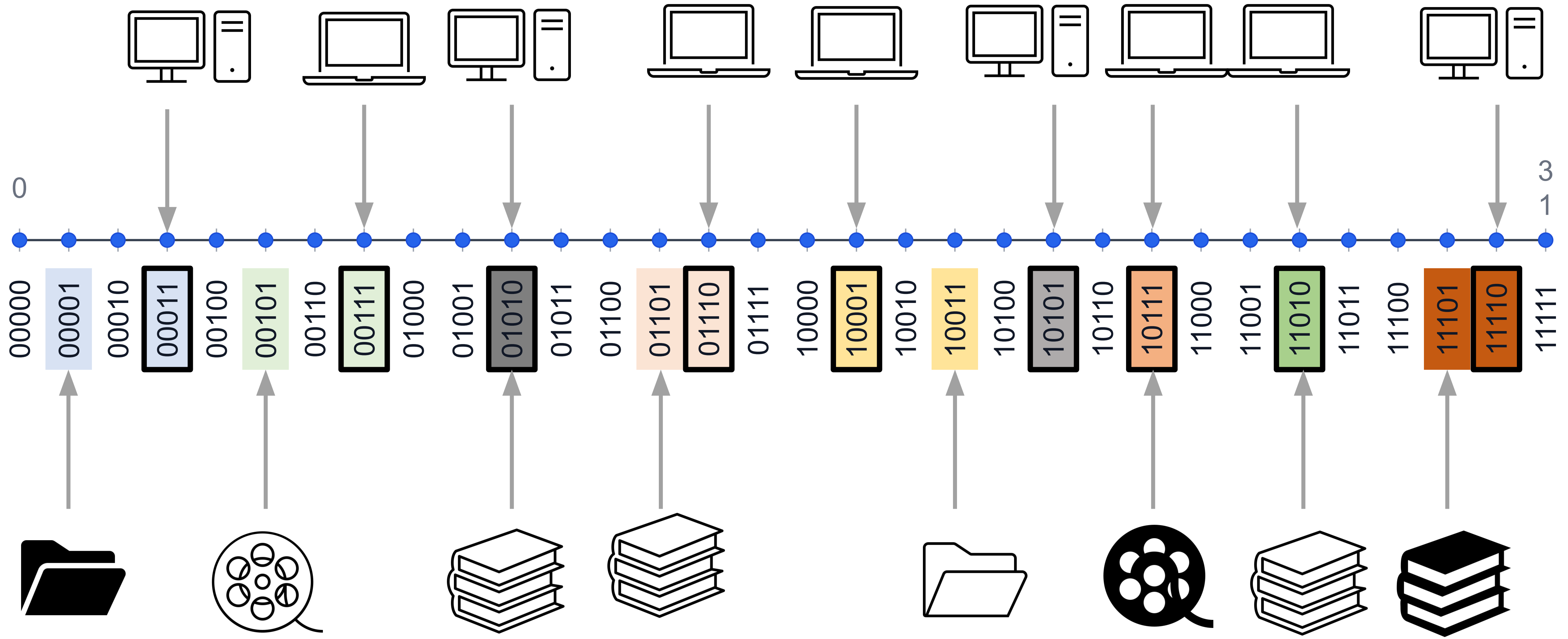
- Distributed dynamic systems
- **Service:** Key-value store using a Distributed Hash Table (DHT)
- **Structure:** Random temporal graphs
- **Several variants:** Kademlia [Maymounkov, Mazières, 2002], Chord, Tapestry.
  - BitTorrent: DHT for trackerless peer discovery
  - InterPlanetary File System (IPFS): Kademlia-based decentralized content routing
  - Ethereum: Kademlia-inspired peer discovery protocol
  - eMule (Kad Network): Early large-scale Kademlia deployment



Crash Course on Kademlia



Crash Course on Kademlia



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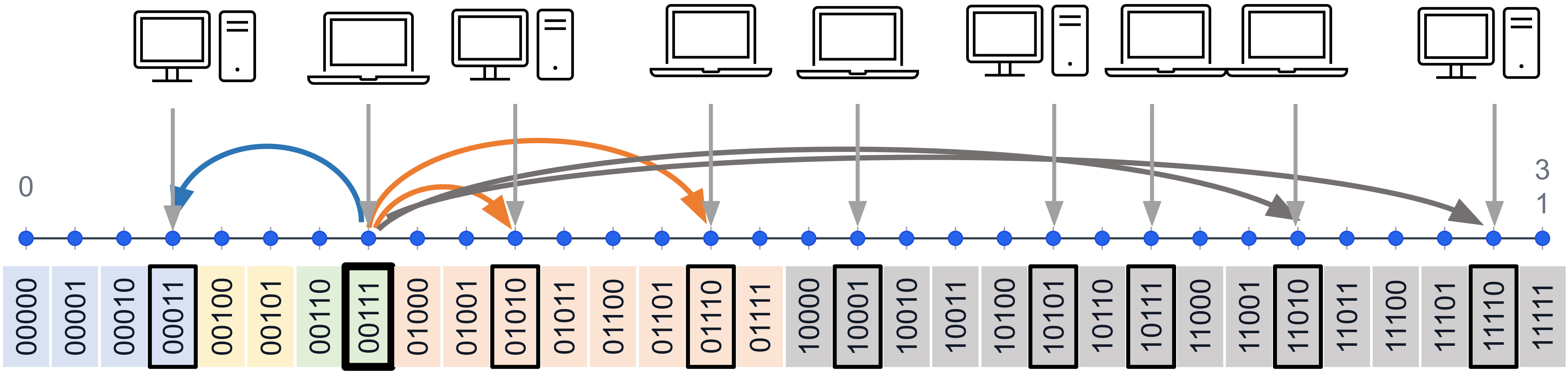
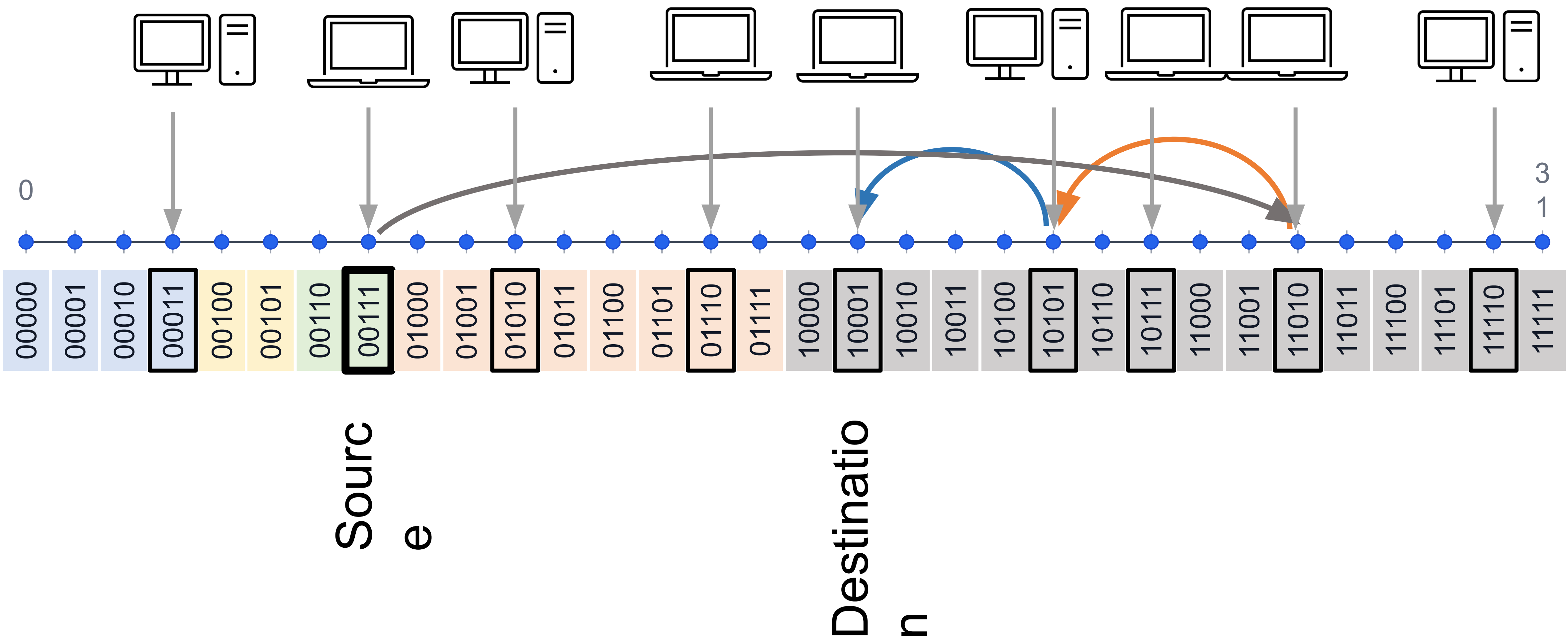


Buckets of 00111:

Common prefix	Peers
0011	
001	
00	00011
0	01010, 01110
-	11010, 11110

- Peers in a bucket are the contacts of a node
- Nodes can appear and disappear
- Buckets contents change over time

Crash Course on Kademlia

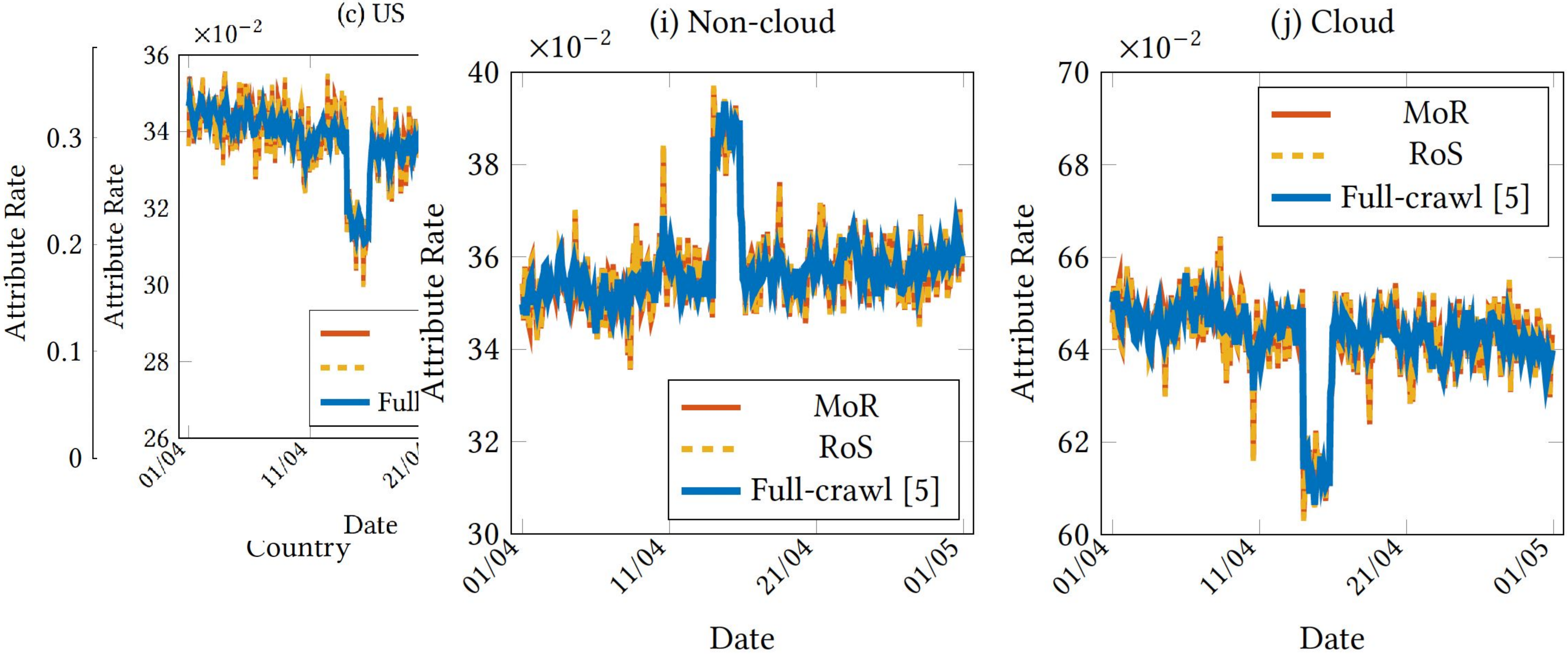


Application: IPFS and Kademlia

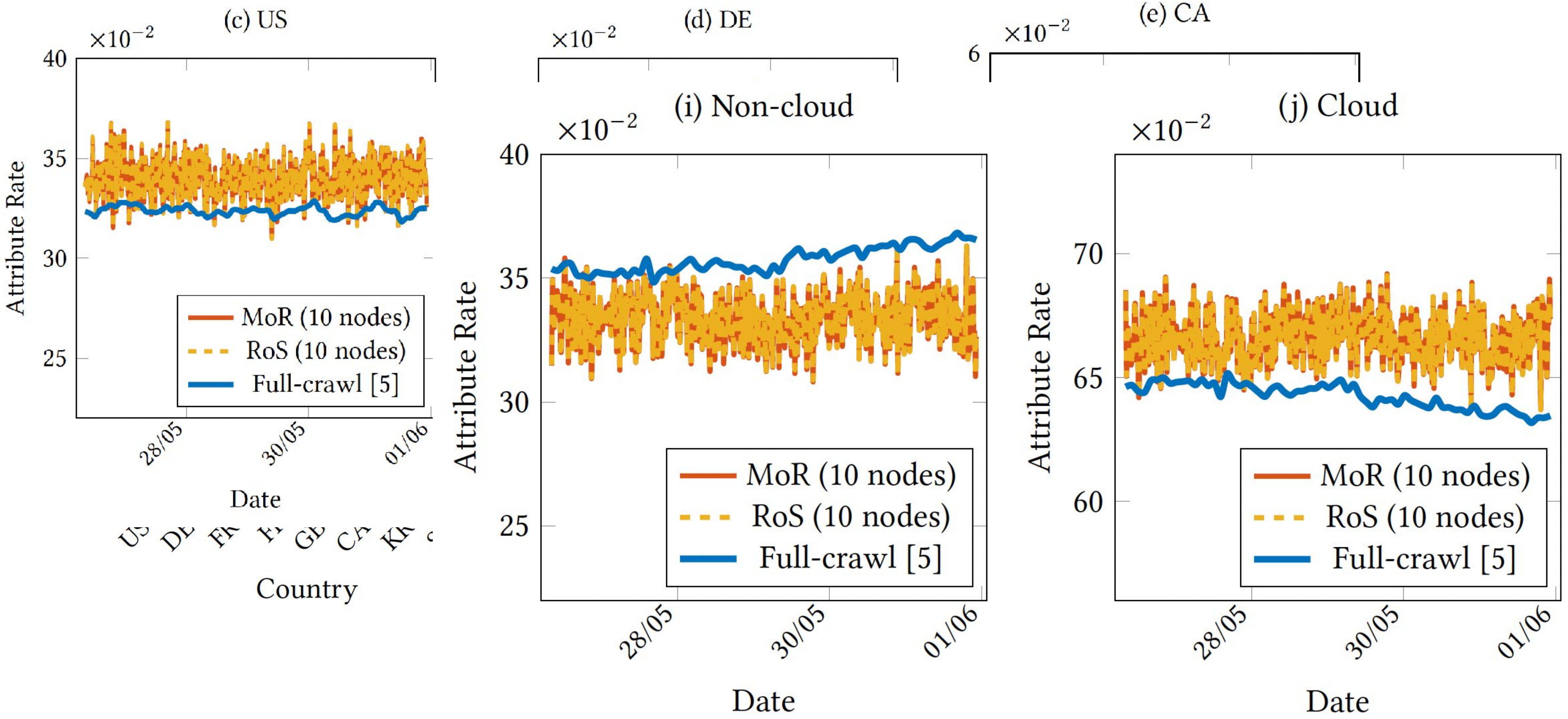
Inter Planetary File System (IPFS):

- Uses a Kademlia DHT for routing
- Implemented with the library LibP2P
- We crawl the entire IPFS DHT 12 times per day between April 1 and May 1, 2026, for a total of 360 snapshots
- We also operate 10 IPFS nodes spread among three servers in Germany, and collect their local buckets
- From the IP address of a node we get a geolocation using the MaxMind GeoLite2 database
- From the IP address of a node we get to a known cloud provider using the Udger database

Results Sampling 10 Random Nodes in Snapshots



Results Sampling 10 Local Nodes



Application: Vote Intention

- We have used NSUM and indirect surveys for vote intention
- Collection of 200 responses in Madrid, Valencia, and Andalucia stratified by age group
- Cleaning to remove outliers

Method	Voting percentage				Seats (D'Hondt method)			
	Vox	PP	PSOE	Sumar	Vox	PP	PSOE	Sumar
GAD3 (July 23)	NA	NA	NA	NA	5	17	10	5
CIS (June 23)	NA	NA	NA	NA	4–5	14–15	10–11	7–8
Naïve (July 23)	16.1	35.1	28.1	16.2	6	14	11	6
MoS (July 23)	14.9	37.8	27.5	16.1	6	15	11	5
MLE (July 23)	15.4	36.3	29.1	16.4	6	14	11	6
PIMLE (July 23)	14.6	39.2	27.6	14.7	5	16	11	5
Real (CERA excluded)	**13.9**	**40.2**	**27.7**	**15.3**	**5**	**15**	**11**	**6**
Real (CERA included)	**13.9**	**40.2**	**27.6**	**15.3**	**5**	**16**	**10**	**6**

Arevalillo, Jorge M., et al. "Network scale-up methods on aggregated relational data to estimate the outcome of elections." *Social Networks* 86 (2026): 312-324.

Conclusions

- Indirect surveys and sampling are a useful tool to monitor networks
- Provide good estimates even with limited number of responses
- Can be easily deployed online
- Can be easily used to monitor trends
- Limits and assumptions that make it applicable have to be explored
- Not widely exploited over time and space: opportunities for research in temporal graphs

Open Problems

- Connect indirect sampling and ARD with temporal graphs
- Model node arrival and departure (burstiness?)
- Model creation and removal of links
- Estimate graph parameters or construct graphs from complete data
- Estimate graph parameters or construct graphs from partial or stale information
- Deal with biases in temporal graphs

References

- Garcia-Agundez, Augusto, et al. "Estimating the COVID-19 prevalence in Spain with indirect reporti surveys." *Frontiers in Public Health* 9 (2021): 658544.
- Srivastava, Ajitesh, et al. "Nowcasting temporal trends using indirect surveys." *Proceedings AI Conference on Artificial Intelligence*. Vol. 38. No. 20. 2024.
- Díaz-Aranda, Sergio, et al. "Error bounds for the network scale-up method." *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 2*. 2025.
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Honorable Mention to Best Paper for the AI for Social Impact track at AAAI-2024

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Thank you!!

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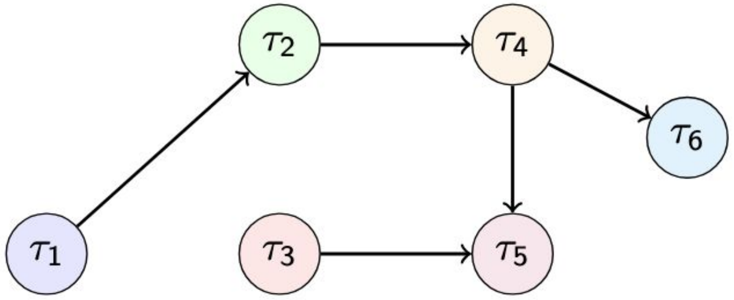
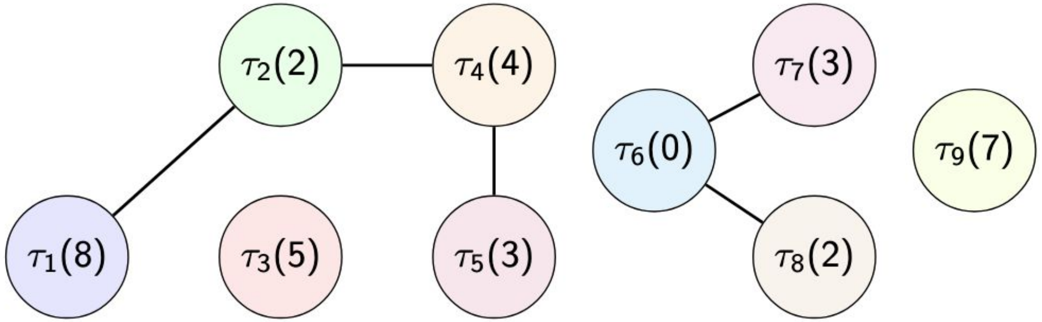
Financiado como parte de la respuesta de la Unión a la pandemia de COVID-19

Parallelization of block construction

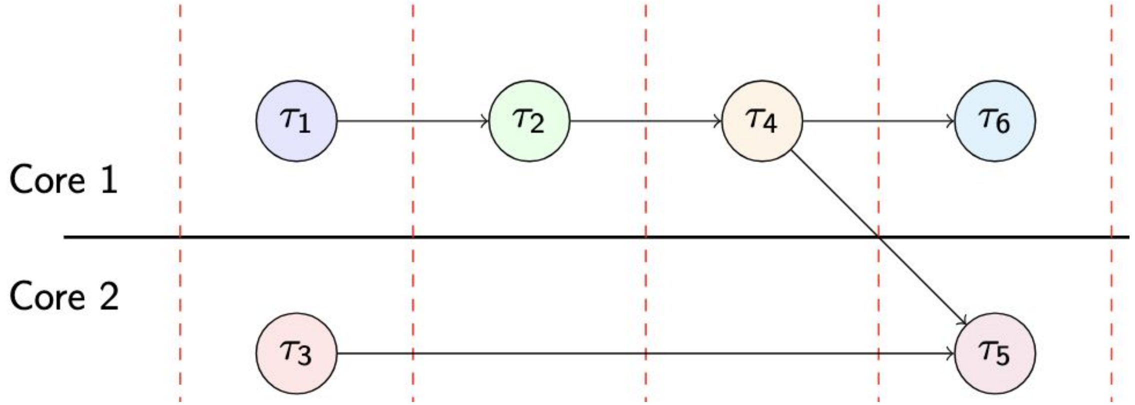


DAG of a block



Transactions in mempool

Parallel Execution:



Parallel Execution: